Discovering the Lyrical Drivers of TikTok Virality

Ruben Koelewijn
s1105300

October 22, 2026

Research Question

The knowledge we wish to extract from the data is which lexical and structural features of current song lyrics drive a song to be TikTok viral [1].

- **Subquestion 1:** Can we train a machine learning classifier to accurately distinguish TikTok-viral songs from standard commercially successful songs based solely on extracted text features?
- **Subquestion 2:** Which interpretable text features hold the highest predictive importance for platform virality?

Motivation

Recent studies show that song lyrics are becoming more simple and repetitive [3]. We hypothesize that this trend is created by short-form video platforms such as Tiktok. As Alaminos Fernández and Alaminos-Fernández [2026] shows, having your song become "viral" on Tiktok is not random, but emerges from a structured convergence of sonic properties, emotional expressivity, and algorithmic dynamics. Previous work by Prevedello et al. [2024] shows that predicting song popularity based on song lyrics is possible by utilizing a Large Language Model. Their approach, however, doesn't explain the lyrical reasons *why* a certain song becomes popular. By using text mining techniques, we aim to find this out.

Data Description

The data we will be analysing is created from a data set which contains roughly 2.56 GB of English song lyrics [2]. Since Tiktok has only existed from 2018, we will focus on songs that have been released between 2020 and 2025 . We will analyse the unstructured text of these songs and label trending TikTok songs according to playlists on Spotify (since TikTok's API is locked). We wish to create a dataset which will consist of 3,000 songs that have been labelled as "Viral" and 3,000 that were "Non-Viral".

Validation

Validation of our findings will be done by creating a 80/20 training and test set to prevent overfitting. Futhermore we will investigate the performance metrics of Accuracy, Precision, Recall, and F_1-scores to identify any biases.

[1] Tiktok virality is a concept defined as "blowing up quickly and engaging followers in a way that is surprising", see https://www.forbes.com/sites/johnbbrandon/article/how-to-go-viral-on-tiktok/

References

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